

COURSE FINAL PROJECT DOCUMENTATION

CASA PARAISO

Spa Service Browsing & Appointment Booking Application



An offline-first academic prototype built with Flutter & Dart for Android and Responsive Web platforms.

Author / Developer: Course Project Student

Platform Stack: Flutter 3.x | Dart | Riverpod | GoRouter | SharedPreferences

Target Platforms: Android (API 26+) & Responsive Web

Date of Defense: August 2026

1. Executive Project Overview & Scope

Casa Paraiso is a dedicated customer-facing mobile and web application designed for a body and wellness spa situated in Barangay Cuta East, Santa Teresita, Batangas, Philippines. The application serves as an interactive catalog and local appointment management tool for spa clientele.

Academic Scope & Prototype Boundaries:

This software application is built as an *offline-first, local-only academic prototype*. All appointment records, user preference settings, customer profiles, and privacy consent choices are maintained exclusively on the user's personal device via `SharedPreferences`. The app does not connect to a live backend server, does not transmit bookings over network channels, and does not check real-time spa seating capacity.

1.1 Primary Core Features

- **First-Launch Onboarding:** A 3-slide visual introductory wizard explaining service discovery, local appointment management, and privacy & local notifications.
- **Service Catalog Browsing:** Interactive catalog displaying four core spa packages (Gaia Touch, Tethys Flow, Hestia Warmth, Aurora Breeze) with pricing, duration, bilingual descriptions, included treatments, and paid extras.
- **4-Step Guided Booking Wizard:** Intuitive step-by-step booking flow:
 1. *Options Step:* Select massage style (Swedish, Shiatsu, Traditional Hilot), included treatment choice, and optional paid add-ons.
 2. *Date & Time Step:* Select business date and valid Manila-timezone 30-minute time slot.
 3. *Customer Details Step:* Input customer full name, validated Philippine mobile number (+639...), optional email, notes, and local storage consent.
 4. *Review & Confirmation Step:* Summary review with total price, duration, and reference code generation.
- **Appointment Management:** Tabbed booking history (Upcoming, Completed, Cancelled) with full record inspection, date/time rescheduling, and explicit cancellation.
- **Local Notification System:** Automated Android local notification scheduling 24 hours and 1 hour before upcoming appointment start times.
- **Bilingual Localization:** On-the-fly language switching between English (default) and Filipino.
- **Official Contact Shortcuts:** Direct phone dialer launcher (DITO, TM, Landline), Facebook page link, Messenger chat link, and Google Maps location launcher.

2. System Architecture & Technology Stack

The project is developed using a layered, feature-first architecture that decouples state management and pure business logic from UI rendering components.

Layer Component	Technology / Package	Architectural Role & Description
Core UI Framework	Flutter 3.x (Dart 3.x)	Cross-platform declarative rendering engine for Android and Web.
State Management	<code>flutter_riverpod</code> (2.x)	Reactive state notifier pattern without code generation; handles dependency injection.
Navigation & Routing	<code>go_router</code>	Declarative hash-based URL routing for web compatibility (e.g. GitHub Pages deep links).
Local Persistence	<code>shared_preferences</code>	Key-value local storage with versioned JSON domain keys and isolated error recovery.
Timezone & Time Rules	<code>timezone</code> package	Asia/Manila IANA timezone database for exact business hour and slot calculation.
Notifications	<code>flutter_local_notifications</code>	Platform-native Android appointment reminder scheduling (24h and 1h prior).
Localization	Flutter ARB + <code>intl</code>	Bilingual translation delegate managing English (en) and Filipino (fil) strings.
External Launcher	<code>url_launcher</code>	Handles deep linking to telephony dialer, web browser, Messenger, and Google Maps.

2.1 High-Level Architecture Diagram

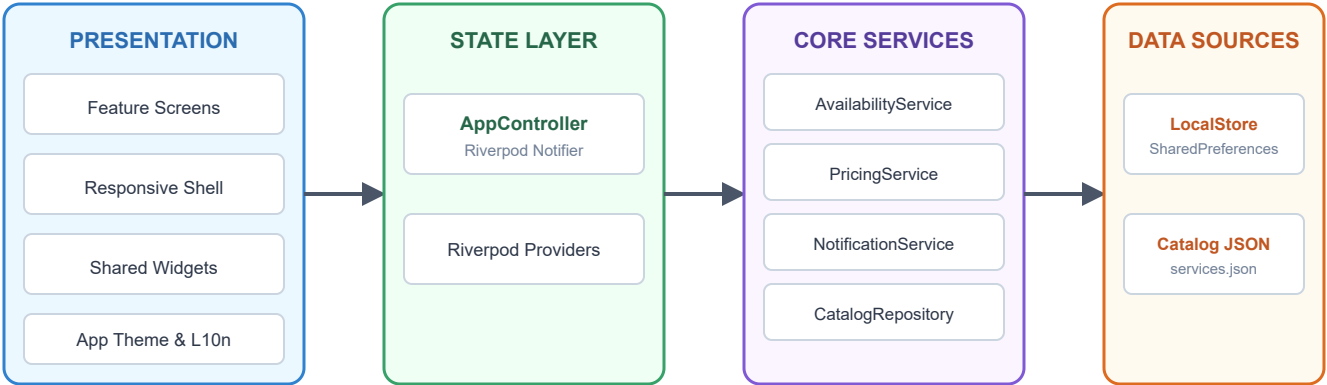


Figure 1: High-Level Layered Architecture & Data Flow

3. Data Models & Catalog Specifications

All domain entities are implemented as immutable Dart classes supporting complete JSON round-trip serialization. Appointments store a full immutable snapshot of the selected package, treatment, extras, and pricing at creation time to maintain accurate history even if static catalog prices are updated later.

3.1 Service Catalog Breakdown

Package Name	Base Price	Duration	Included Treatment Options (Choice of 1)
Gaia Touch	₱499.00	60 min	None (Signature massage only)
Tethys Flow	₱649.00	60 min	Ventosa or Hot Compress
Hestia Warmth	₱749.00	90 min	Ventosa, Hot Stone, or Hot Compress
Aurora Breeze	₱849.00	120 min	Ventosa, Hot Compress, or Hot Stone

3.2 Massage Styles & Paid Add-Ons

- **Massage Styles (Included with every package):** Swedish, Shiatsu, or Traditional Hilot.
- **30-minute Back Massage (Extra):** ₱298.00 (+30 minutes added to total appointment duration).
- **VIP Room (Extra):** ₱200.00 (Price increment only; 0 added minutes).

3.3 Appointment Class Structure

```
class Appointment {
  final String id;           // Unique microsecond timestamp string
  final String reference;    // Format: CP-YYYYMMDD-XXXX (e.g. CP-20260828-AB3K)
  final ServicePackage service; // Full package snapshot
  final String massageStyle; // Swedish / Shiatsu / Traditional Hilot
  final String? treatment;   // Selected included treatment
  final List<PaidExtra> extras; // List of selected paid extras
  final DateTime startUtc;    // Appointment start instant in UTC
  final DateTime endUtc;     // Appointment end instant in UTC
  final double total;        // Precomputed total price in PHP
  final CustomerProfile customer; // Customer name, mobile (+63), optional email
  final DateTime createdAtUtc; // Creation timestamp in UTC
  final String? notes;       // Free-text note (max 250 chars)
  final AppointmentStatus status; // confirmed | cancelled
}
```

4. Business Logic & Availability Rules

Appointment slot availability is computed dynamically by `AvailabilityService` using pure Dart logic independent of device timezone settings.

4.1 Business Hour & Timezone Constraints

- **Fixed Business Timezone:** `Asia/Manila` (UTC+8).
- **Daily Spa Operating Hours:** 1:00 PM (13:00) until 1:00 AM the following calendar day (25:00).
- **Slot Interval:** Starts are generated in 30-minute increments.
- **Duration Cutoff:** The calculated total duration (package duration + extras duration) must fit completely prior to the 1:00 AM closing limit.
- **Lead Time:** Required minimum start time lead of at least **1 hour** from the current Manila time.
- **Booking Horizon:** Maximum forward booking limit of **30 days**.
- **Conflict Prevention:** Slots that overlap with the customer's existing *confirmed* appointments are filtered out. During rescheduling, the appointment being modified is excluded from conflict checks.

Reference Code Generation:

Reference codes follow the format `CP-YYYYMMDD-XXXX` where `XXXX` is a cryptographically secure 4-character alphanumeric string generated from non-ambiguous uppercase characters. The service performs up to 20 collision checks against existing local records.

5. Persistence & Error Isolation

Local persistence utilizes Flutter's `SharedPreferences` plugin with distinct versioned keys for each data domain.

Domain Key	Data Type	Purpose & Storage Scope
<code>onboarding.v1</code>	Boolean	Stores onboarding completion state; prevents re-showing slides after first run.
<code>locale.v1</code>	String	Persists selected language code ('en' or 'fil').
<code>profile.v1</code>	JSON String	Stores reusable <code>CustomerProfile</code> (name, normalized mobile, email).
<code>appointments.v1</code>	JSON Object	Contains versioned array of <code>Appointment</code> snapshots.
<code>consent.v1</code>	Boolean	Records user acceptance of local data storage policy.

Fault-Tolerant Recovery Mechanism:

If a stored JSON string for profile or appointments becomes malformed or corrupted due to storage interruption, `LocalStore.load()` catches the parsing exception, clears *only* the malformed key, preserves all unaffected keys, and flags `recoveredData = true` in `AppState` to display a non-disruptive notification snackbar to the user.

6. Brand Experience & Design System

The visual aesthetic of Casa Paraiso embodies a light-only Material 3 design system inspired by a calm, tropical wellness sanctuary.

6.1 Brand Palette & Design Tokens

- **Cacao (Primary - #7A3518)**: Rich warm reddish-brown used for primary buttons, selected step bars, and main headlines.
- **Palm (Secondary - #2F5D45)**: Deep foliage green for icons, badges, leaf marks, and secondary accents.
- **Brass (Tertiary - #A98245)**: Muted warm brass accent color.
- **Sand (Border/Pills - #E4D4BD)**: Soft warm beige for borders, container dividers, and chip backgrounds.
- **Cream (Scaffold Background - #FBF7EF)**: Off-white main background tone.
- **Surface (Card - #FFFDF8)**: Crisp elevated card surface background.

6.2 Dual Typography Hierarchy

- **Cormorant Garamond (Serif)**: Used for large display titles, section headers, hero headlines, and emotional brand copy.
- **Manrope (Sans-Serif)**: Used for navigation items, forms, body text, prices, timestamps, and interactive controls.

6.3 Accessibility & Responsive Standards

- Minimum touch target size of 48dp (all primary buttons formatted at 52dp height).
- WCAG AA color contrast compliance across text elements.
- Responsive navigation switching: Bottom `NavigationBar` on mobile screens (< 600dp width) and side `NavigationRail` on tablet/desktop views (≥ 600dp).
- Content container width constrained to 1120dp (760dp for detail views) to prevent uncomfortable horizontal stretch on wide desktop screens.

7. Quality Assurance & Verification Results

The project codebase undergoes rigorous static analysis and automated unit, widget, and integration testing.

Test Suite File	Tests Passed	Key Coverage Scope
test/booking_services_test.dart	Passed (4/4)	Pricing/duration addition, Philippine mobile normalization, cross-midnight slot generation, lead time & horizon enforcement, overlap detection, rescheduling exclusion.
test/local_store_test.dart	Passed (4/4)	Local JSON serialization, SharedPreferences round-trips, corrupt payload error recovery, domain key isolation.
test/widget_test.dart	Passed (1/1)	Onboarding flow completion, 4-destination navigation switching, responsive shell layout.

7.1 Automated Suite Output

```
$ flutter analyze
Analyzing Casa_Paraiso_FlutterDart...
No issues found! (ran in 2.5s)

$ flutter test
00:01 +9: All tests passed!
```

8. Conclusion & Future Roadmap

The Casa Paraiso mobile and web application delivers a complete, dependable, and aesthetically elevated appointment booking experience tailored for wellness clients. The architectural decoupling of presentation features, state controllers, and core domain services ensures that future integration with remote production backends (such as REST/GraphQL APIs, remote authentication, or real-time capacity database synchronizers) can be performed without refactoring the user interface layer.